



Indian Institute of Technology Kharagpur

Class Test II 2024-25

Date of Examination: 11 Feb, 2024

Duration: 30 Minutes

Subject No.: CS60050

Subject: Machine Learning

Department/Center/School: Computer Science

Credits: 3

Full marks: 30

Name: _____

Roll Number: _____

Question:	1	2	3	Total
Points:	10	10	10	30
Score:				

Write the answers in the space provided after the questions only.

1. (10 points) In what range will the value of the coefficient α_i for a training datapoint x_i lie for a hard-margin SVM in the following situations? Give a short 1-sentence reason:

- (a) When x_i is on the correct side of margin.

Solution:

$$\alpha_i = 0$$

- (b) When x_i is on the margin.

Solution:

$$0 < \alpha_i$$

- (c) When x_i is on the wrong side of the margin.

Solution:

It is not possible to have a training datapoint on the wrong side of the margin.

2. (10 points) Consider the following dataset for a 2-class classification problem:

Class	0	0	0	1	1
Gender	M	F	F	F	M
Age	25	22	28	32	38

Calculate the class conditional densities considering that Gender is a Bernoulli Random Variable and Age is a Gaussian Random variable.

What is the minimum age of a male for him to be classified as class 1.

Solution:

$$\mu_0 = 25, \sigma_0^2 = 6$$

$$\mu_1 = 35, \sigma_1^2 = 9$$

$$P_0(M, F) = 1/3, 2/3$$

$$P_1(M, F) = 1/2, 1/2$$

For boundary, age A should be:

$$\frac{1}{\sqrt{2\pi(9)}} e^{-(A-35)^2/18} * \frac{1}{2} * \frac{2}{5} = \frac{1}{\sqrt{2\pi(6)}} e^{-(A-25)^2/12} * \frac{1}{3} * \frac{3}{5}$$

$$-\frac{(A-35)^2}{18} = -\frac{(A-25)^2}{12} + \ln\left(\sqrt{\frac{3}{2}}\right)$$

3. (10 points) Consider the following algorithm for estimating bias and variance of a model $y(x; \theta)$ using bootstrap samples:

- 1: Input datasets $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$ and $D_{test} = \{(x_1, y_1), \dots, (x_m, y_m)\}$
- 2: Compute bootstrap datasets $B_1, \dots, B_K$ from D .
- 3: Train models $\theta_1, \dots, \theta_K$ from the bootstrap samples $B_1, \dots, B_K$
- 4: Compute the predictions $\hat{y}_i(k)$ _____
- 5: Compute the average predictions $\bar{y}_i$ _____
- 6: Compute Bias = _____
- 7: Compute Variance = _____

Complete the algorithm by filling in the blank spaces with the exact formulae for for each of the quantities.

Solution:

- 1: Input datasets $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$ and $D_{test} = \{(x'_1, y'_1), \dots, (x'_m, y'_m)\}$
- 2: Compute bootstrap datasets $B_1, \dots, B_K$ from D .
- 3: Train models $\theta_1, \dots, \theta_K$ from the bootstrap samples $B_1, \dots, B_K$
- 4: Compute the predictions $\hat{y}_i(k) = y(x'_i; \theta_k) \forall i = 1, \dots, m \forall k = 1, \dots, K$
- 5: Compute the average predictions $\bar{y}_i = \text{Avg}_k(\hat{y}_i(k)) \forall i = 1, \dots, m$
- 6: Compute $\text{Bias} = \frac{1}{m} \sum_{i=1}^m (y'_i - \bar{y}_i)^2$
- 7: Compute $\text{Variance} = \frac{1}{m} \sum_{i=1}^m \left[\frac{1}{K} \sum_{k=1}^K (\hat{y}_i(k) - \bar{y}_i)^2 \right]$

Rough space

